Computational Fluid Dynamics
Project based Learning

Topic: Verification and Validation of Turbulent Flow around a Clark-Y Airfoil

Introduction

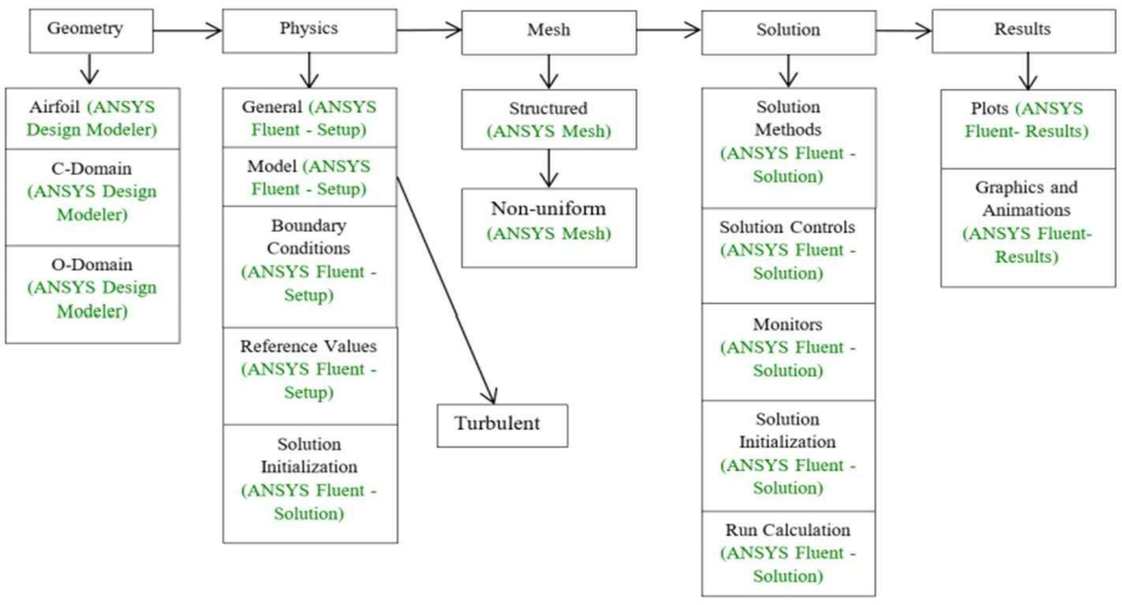
Clark Y is the name of a particular airfoil profile, widely used in general purpose aircraft designs, and much studied in aerodynamics over the years. The profile was designed in 1922 by Virginius E. Clark.

For many applications the Clark Y has been an adequate airfoil section, it gives reasonable overall performance in respect of its lift-to-drag ratio. The Clark Y has found favour for the construction of model aircraft, The Clark Y is appealing for its near-flat lower surface, which aids in the construction of wings on plans mounted on a flat construction board.

Fig. Clark Y Airfoil

Objective

The Purpose of this project is to simulate turbulent airfoil flows following "CFD process" by an interactive step-by-step approach and to conduct verifications. using ANSYS to conduct verification for lift coefficient and pressure coefficient distributions, and validation for pressure coefficient distribution, including effect of numerical scheme.



Flow Chart for "CFD Process" for airfoil flow

Simulation Design

The problem to be solved is that of turbulent flows around a Clark-Y airfoil. Reynolds number is 143,000 based on the inlet velocity and airfoil chord length. The figure 1 show the illustrations for C type and O type domains (Note: the figures are not in the exact scale as the true size of the domain and airfoil).

Boundary conditions for C type of mesh will be "inlet", "outlet", "symmetry" and "airfoil", as described later. Boundary conditions for O type of meshes will be "inlet", "outlet", and "airfoil". Uniform flow was specified at inlet. For outlet, zero gradients are fixed for all velocities and pressure is constant. No- slip boundary condition will be used on the "airfoil". Symmetric boundary condition will be applied on the "symmetry".

Results

Table 1 - Main particulars

Parameter	Symbol	Unit	O-type	C-Type
Chord Length	C	m	0.3048	0.3048
Downstream length	Lo	m	-	5
Radius	Rc	m	5,4,3,2,1	5
Angle of attack	α	degree	0,6	0

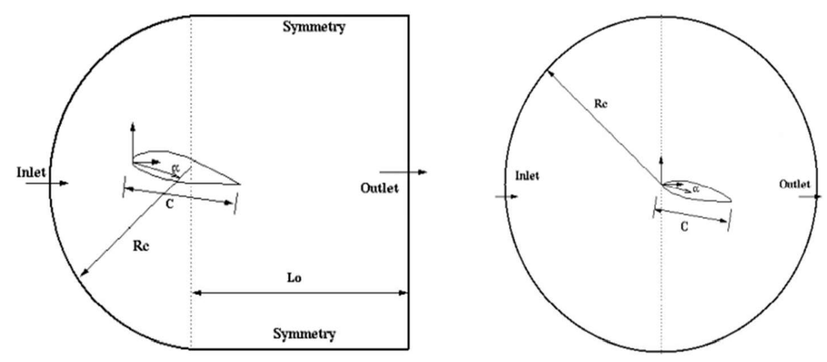


Figure 1 – C and O domain shapes and boundary conditions

Results

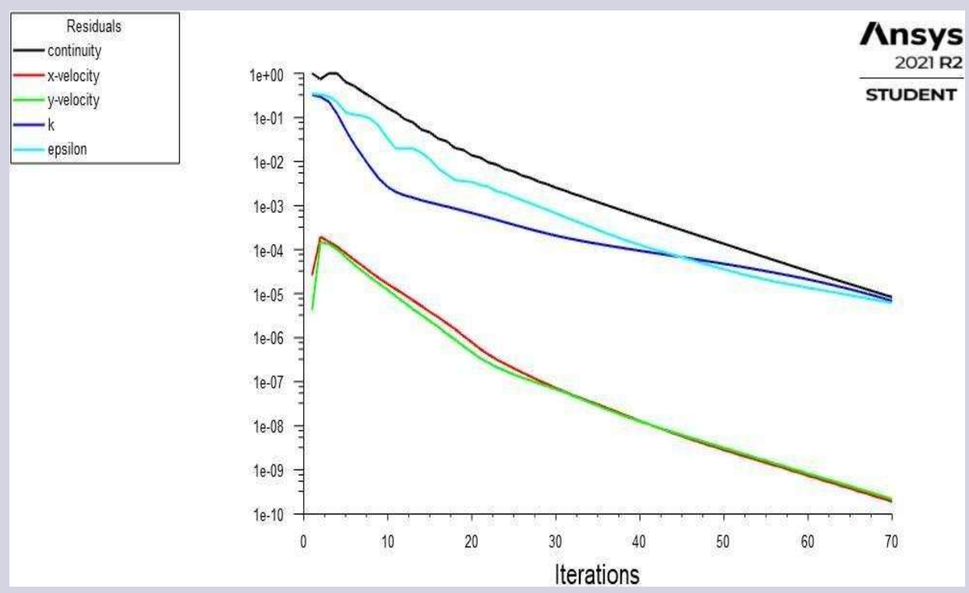


Fig. Residual Plots for an Airfoil

Results

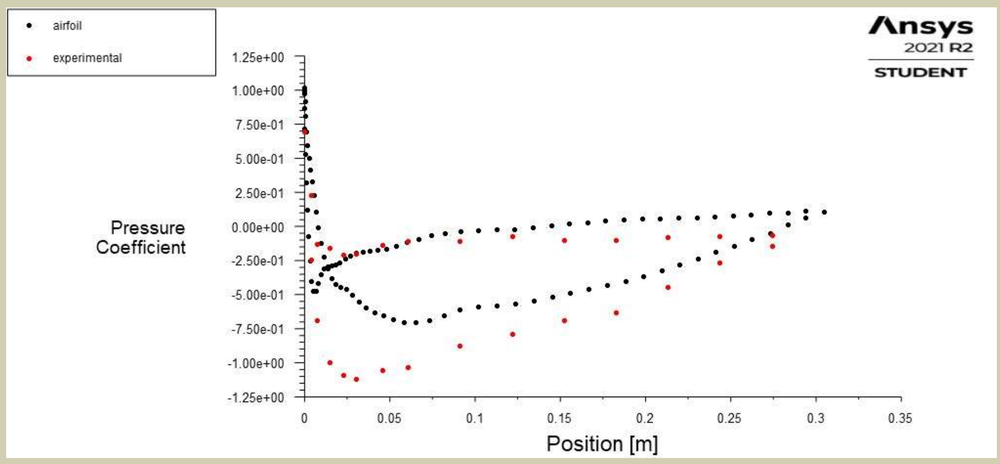


Fig. Pressure Coefficient on Airfoil Surface

Results

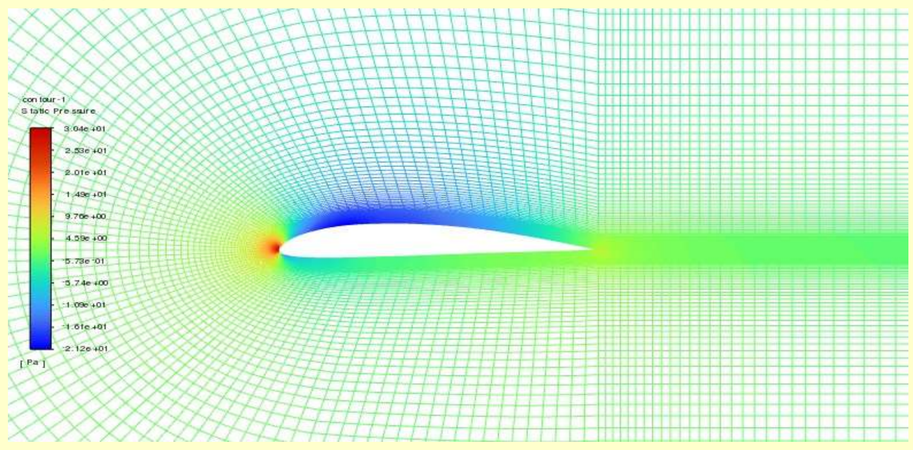


Fig. Pressure contour across Airfoil

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